

Philosophy and Mathematics.

A Propaedeutic Treatise

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The author of the present treatise, who neither is nor wishes to be a mathematician, has found in philosophy, the general theory of science, an incitement to learn the language of mathematics as well, and, with respect to its leading fundamental concepts, so far as possible, to enter into its way of thinking.

I.

Mathematical Functions.

The word “function” occurs in a very wide sense. Wherever there can be talk of acting, suffering, doing, function denotes a particular law-governed activity. It is said of an official¹, that he is in function; one speaks of the function of the eye, of the function of a nerve, and so forth.

On the domain of pure magnitudes there can, as regards these, just as little be talk of real doing as of real suffering: an a cannot act, a b cannot suffer, an x cannot function in the likeness either of an official or of a nerve.

Since nevertheless the designation “function” is not only retained, but even held fast as a concept in mathematics, one must be able, in the mathematical application, to point out something corresponding to the fundamental meaning.

To be sure, pure magnitudes cannot exert real activity, but by being placed in different connections they can nevertheless essentially alter their mutual relations. A magnitude can, so to speak, be appointed to different offices; it can, as an addend ($a + x$), as a factor (ax), as an exponent of a power (a^x , x^a), as a logarithm, and so forth, acquire a very different influence, and to that extent appear in different functions.

Here, however, is another side of the concept, which must be brought out especially in the theory of magnitude. One speaks not merely of functioning, of being in function; one speaks also of that which is brought about by the functioning activity. Outside mathematics one thinks, in connection with function, chiefly of the activity, in mathematics chiefly of the result; outside mathematics the point is to be in function at something, within mathematics to be a function of something. The law-bound relation between a dependent and an independent shows itself, however, on both sides.

“If a magnitude y depends, according to a certain law, on other magnitudes x , z , t , u ..., so that every value of y is determined by the values arbitrarily assigned to x , z , t ,

¹With categorical emphasis Chamberlain Norden in “Gamle Minder” stresses his “term of function.”

$u \dots$, then y is said to be a function of these magnitudes, all regarded as variable, y dependent, the others independent. This is commonly denoted: $y = f(x, z, t, u \dots)$ ².

According to this mathematical determination of the concept, changes will thus proceed from the independents and exert their lawful influence upon the dependent. In order to recall the likeness between non-mathematical and mathematical functions, one might indeed regard the independents $x, z, t, u \dots$ as the functioning ones, the agents, that cause the changes in the suffering y . But since the conception is figurative, one likewise sees why $x, z, t, u \dots$ are not said to be in function with respect to y , whereas y is expressly said to be a function of $x, z, t, u \dots$

In the function, then, the magnitudes are variable; but magnitudes can of course also be invariable. That every magnitude must be equal to itself, and that every magnitude can, by increase or decrease, at every moment become unequal to itself, is a contradiction that is resolved in the concept of function. In and for itself no magnitude is either invariable or variable, for no magnitude is anything alone in and for itself. Whether the particular magnitude, on the other hand, in its relation to other magnitudes, is to play the role of the invariable or of the variable, and, within variability, again that of the dependent or of the independent, is determined by the function, and in turn determines the function.

If the concept function is now, in the first place, determined by the concept of dependence, then it depends again on determining the mode of dependence, that is to say: on pointing out the law.

In the mathematical presentation of the different functions there shows itself a scale of case, rule, and law, which must become an object of philosophy's attention.

$$\text{That } 2(3 + 4) = 2 \cdot 3 + 2 \cdot 4 = 6 + 8 = 2 \cdot 7 = 14,$$

$$\text{that } (3 \cdot 4)^2 = 12^2 = 3^2 \cdot 4^2 = 9 \cdot 16 = 144,$$

$$\text{that } 2^3 \cdot 2^4 = (2 \cdot 2 \cdot 2) \cdot (2 \cdot 2 \cdot 2 \cdot 2) = 2^{3+4} = 2^7,$$

can be experienced by trial, and seen by comparison. In this way one can test various cases and abstract a rule from them. But, however important the rule may be for practical use, it is nevertheless not, like the law, a necessary expression of the essence of the matter.

²C. Ramus. *Algebra og Functionslære*. Kjøbenhavn 1840, S. 14.

In order to obtain a rule as comprehensive as possible, one must test as many cases as possible; but even if one tested cases by the thousand without meeting a single exception, the rule could not thereby be raised to a law.

The rule expresses only *that a relation is*, not that it necessarily must be; but the law holds as law in virtue of the concept. That, when the law is to be known, it does not depend on the number of cases, but on *the essence of the matter*, and hence on insight, can be seen from the following example.

That $\frac{1}{1-x} = 1 + x + x^2 + \dots$ will yield an infinite series is proved not by continuing the division to infinity, but by understanding what the division, on the given presupposition, necessarily brings with it.

How easily a series of cases falling under a rule can deceive, when one would from this infer a law, is well known. One seeks, for example, a law for the prime numbers, and finds: $2^3 - 1 = 7$, $2^5 - 1 = 31$, $2^7 - 1 = 127$, all prime numbers. But if one now, from these three cases, would draw the conclusion that $2^{2m-1} - 1$ must necessarily be a prime number, one would badly miscalculate, since already the fourth case, $2^9 - 1 = 511 = 7 \cdot 73$, gives no prime number.

That in the cited example the exception to the rule already occurs with the fourth case is, in the whole consideration, itself as it were a matter of chance. Had the exception occurred with the 99th case, we should still just as little have had any law, but should on the contrary have been more exposed to being deceived by the multitude of cases³.

That mathematics, instead of definite numerical values, employs general designations — instead of

$$\begin{array}{ll} 2 \cdot (3 + 4) = 2 \cdot 3 + 2 \cdot 4 & a(x + y) = ax + ay; \text{ instead of} \\ (3 \cdot 4)^2 = 3^2 \cdot 4^2 & (xy)^a = x^a y^a; \text{ instead of} \\ 2^3 \cdot 2^4 = 2^{3+4} & a^x a^y = a^{x+y} \end{array}$$

— is, as regards the cognition of laws, an important step. The understanding, which from the beginning was entangled in the consideration of the definite values, the diverse cases, the individual rules, raises itself, by means of the general designations,

³Fermat supposed that $2^{(2^n)} + 1$ was a prime number; but Euler showed that $2^{32} + 1$ is divisible by 641.

to insight into the essence of the relations, and thereby to insight into the law.

How intimately the law hangs together with the concept can be seen by a glance at the logarithmic function. $a^z = x$, $z = \log x$; $a^t = y$, $t = \log y$, cannot be established by any proof. Here there can be talk neither of case nor of rule nor of law; here it is a question of a determination of the concept for the logarithm, a statement of its category. But once the category of the logarithm is fixed, then there holds as law in virtue of the concept: $a^z a^t = a^{z+t} = xy$, and $z + t = \log(xy) = \log x + \log y$.

In so far as the concept of function everywhere points to definite relations of dependence, we are directed by the functions to perform intellectual, law-governed actions. How intimately these actions fuse with the concept can be seen from the so-called fundamental equations.

As the first of all functions, $a+x$ cannot indeed itself be defined by any fundamental equation; but the four single functions ax , x^a , a^x , and $\log x$ can, on the other hand, be regarded as entirely determined in their nature⁴ by the following four equations.

$$f_1(x+y) = f_1(x) + f_1(y); \quad f_1(x) = ax, \quad \text{for } a(x+y) = ax + ay.$$

$$f_2(xy) = f_2(x) f_2(y); \quad f_2(x) = x^a, \quad \text{for } (xy)^a = x^a y^a.$$

$$f_3(x+y) = f_3(x) f_3(y); \quad f_3(x) = a^x, \quad \text{for } a^{x+y} = a^x a^y.$$

$$f_4(xy) = f_4(x) + f_4(y); \quad f_4(x) = a \log x, \quad \text{for } a \log(xy) = a \log x + a \log y.$$

If we now see in these fundamental equations a demonstration of laws, a presentation of concepts, the difference between mathematics and philosophy will be conspicuous. What is peculiar makes itself known in the language.

For a philosophical presentation of the concept the word is used, and the word sounds different in the different vernaculars; but mathematics is not content with the word; independent of the diversity of the vernaculars, the exact science has its own language: $f(x+y) = f(x) f(y)$ is neither Danish nor French, but mathematical. That the notation which mathematics employs is now also really a language in a stricter sense than, for example, the notation of notes in music, is seen from the fact that it is not confined to indicating different operations of calculation, but expresses oppositions, draws conclusions, and points out the difference between the possible

⁴Jvfr. Ramus: Algebra og Functionslære S. 14–18.

and the impossible.

In the inverse functions we have expressions both for opposition and for conclusion. If the function φ is the inverse of ψ , i.e. $\varphi(\psi(x)) = x$, then also $\psi(\varphi(x)) = x$, i.e. the function ψ is the inverse of φ . If $\varphi(x) = x + a$, then $\psi(x) = a - x$; if $\varphi(x) = ax$, then $\psi(x) = \frac{x}{a}$; if $\varphi(x) = x^n$, then $\psi(x) = \sqrt[n]{x}$; if $\varphi(x) = a^x$, then $\psi(x) = \log x$; if $\varphi(x) = \cos x$, then $\psi(x) = \arccos x$.

The last designation, which indicates a relation between the trigonometric line $\cos x$ and the arc corresponding to it, $\arccos x$, expresses the opposition in such a way that φ and ψ do not signify two mutually cancelling actions; for x is opposed to x in that in $\varphi(x)$ it is an arc, in $\psi(x)$ a straight line.

That mathematics, in its language, can express both the merely possible and the in-itself impossible, shines forth from the manner in which the relation between the rational and the irrational, the real and the imaginary magnitudes is indicated: $a = \frac{b}{c}$ is, for thought, a reality, $b = \pm \sqrt{a^2 - x^2}$ a possibility, and $\sqrt{-1}$, $\sqrt[2m]{-a}$ an impossibility.

While philosophy, then, together with the other sciences⁵, is referred to a grammatical vernacular, mathematics avails itself essentially of a pasigraphic sign-language: this difference of language must surely have its deeper ground. In so far as a language expresses thoughts and concepts, it expresses how the universal and the individual are separated in abstractions and united in judgments, distinguished in judgments and connected in conclusions.

In the grammatical language the universal has a one-sided preponderance over the individual; in the mathematical language the universal is determined so punctually that it can coincide with the single and meet the individual. In the grammatical language the concept assumes a logical form; in the pasigraphic sign-language a mathematical form.

Logic says: “ A is A ”, “every thing is what it is”, and thereby shows that in the single it looks only at the universal.

Mathematics posits $A = A$, and thereby shows that in the universal it expressly looks at the single.

In the logical proposition “ A is A ”, A is like an image that represents “every thing”, but in itself is without significance. In the designation of the single, one does not have

⁵Incomplete sign-systems, such as, for example, the chemical, cannot be placed on the same level as the mathematical.

this single, but only the universal: the word and the thing, the designation and the designated fall apart from one another.

In the mathematical proposition $A = A$, A also indeed represents any magnitude whatsoever, but, inasmuch as A is an example of any magnitude whatsoever, B , C , D ..., it, as a magnitude, at the same time represents itself; $A = A$, for $2 = 2$, $3 = 3$: the universal is posited punctually, the sign and the designated are congruent.

Logic transforms the equation $A = A$ into a grammatical proposition; mathematics transforms the proposition A is A into an equation. Already in this is seen an essential difference between the exact and the dialectical science.

In the proposition as equation there is no difference between subject and predicate. This holds not only of the so-called identical equations, such as $A = A$, but of any equation whatsoever. For what stands on the one side of the equals-sign represents the subject, and what stands on the other side, the predicate; but for every equation it is quite indifferent from which side it is read. If $f(x + y) = f(x) + f(y)$, then of course also $f(x) + f(y) = f(x + y)$. In the mathematical language the predicate is congruent with the subject, for mathematics is an exact science.

If, on the other hand, the equation is transformed, in the grammatical language, into a logical proposition, then an essential difference arises between subject and predicate; for which reason these two components cannot indeed be united by the mere equals-sign, but must be joined by the copula. Now, when the difference between the subject and the predicate in the logical judgment is brought to consciousness, there appears the much-discussed contradiction, that the single is said to be the universal⁶. In the logical language the predicate's being one with the subject is a contradiction, and philosophy a dialectical science.

The mathematical sign-language and the grammatical proposition-language are, in their origin, both natural to sound human understanding, and therefore enter into various combinations.

Mathematical determinations of relation are so far at home in existence that daily life consumes a by no means insignificant quantum of conscious and unconscious mathematics for practical need. Expressions such as "heavy, heavier, quick, quicker,

⁶Already the Megarian Stilpo attacked the logical judgments. He maintained that one could say nothing but identical, or rather tautological, propositions, for example "Man is man," "Horse is horse"; but not "The horse runs," for "horse" and "to run" are not identical; the latter predicate one ascribes also to dogs and lions. Jvfr. Paul Møllers Forelæsninger over den ældre Philosophies Historie.

warm, warmer” still fall outside the mathematical; but, when it is expressly asked: “how heavy?” “how much heavier?” and so forth, the answer can scarcely lack a mathematical element.

The peculiar relation of mathematical thinking to the general-logical shows itself perhaps most strikingly when mathematics itself takes the word in order to pose a problem. Example. “A water-reservoir can be filled by the uniform inflow of several pipes, n in number, and the times in which it is filled by each pipe separately are given: $a_1, a_2, a_3 \dots a_n$. In what time (x) will it be filled, when the inflow takes place through all of them simultaneously?⁷”

He who has not learned mathematics will understand what is here said, and yet not understand it. In its logical fundamental features the matter is not only universally intelligible, but can even become so clear that the problem is solved in the same moment it is posed.

For if one has four pipes of such a kind that a uniform inflow through each of them separately can fill a certain water-reservoir in one hour, then sound human understanding naturally grasps, involuntarily, that the same water-reservoir must, by uniform inflow through all four pipes at once, be filled in a time four times as short, that is, in the course of a quarter of an hour. Here the mathematical merges into the general-logical.

But how different this mathematical is nevertheless from the general-logical is also evident from the cited example. One need only make the relation between the times in which the water-reservoir is filled through each pipe separately a little complicated; and at once the problem is so peculiar that the stiffest logician, the most acute dialectician, who does not know mathematics, racks his brains in vain to solve it.

He who understands the mathematical language finds the problem easy, indeed for every case essentially equally easy; for the equation $\frac{x}{a_1} + \frac{x}{a_2} + \frac{x}{a_3} + \dots \frac{x}{a_n} = 1$, which gives $x = \frac{1}{\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots \frac{1}{a_n}}$, fits all possible cases. This merging of the universal into the individual cases is decisive for mathematical cognition.

In the logical language the formula is a conclusion, in the mathematical the conclusion is a formula. In the formal conclusion the validity of the concept is indeed

⁷C. Ramus. *Elementær Algebra* S. 179.

seen, but always in such a way that the single and the particular vanish into the universal. In the formula the universal is punctualized, and thereby fitted for the exact determination of the particular and the single.

From the well-known syllogism:

“All men are mortal,
Cajus is a man.
Therefore: Cajus is mortal.”

one obtains, although Cajus is expressly named, nevertheless no particular information about Cajus as Cajus. In the minor premise “Cajus is a man” the universal has already so far swallowed up the single that one gains absolutely nothing by inserting into the syllogism either Brutus or Pompejus or Cnejus instead of Cajus.

From the formula $x = \frac{1}{\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n}}$, on the other hand, one obtains, by inserting for $a_1, a_2, a_3 \dots$ a multitude of arbitrary values, the most diverse cases. And yet the formula is universally valid, invariable; only the individual cases are variable. By the formula, then, there is brought about a peculiar agreement between the universal and the single, the invariable and the variable.

With the logical formula-business, at the present stage of the sciences, things stand but poorly; with the mathematical formulas the matter has another bearing. Against the logical formula-business there is a scientific remedy: a sound and sharp dialectic; against the mathematical formula-business no remedy is needed, for insight into the real laws is conditioned by insight into the mathematical formulas⁸.

The scientific character in the development of these formulas can already be seen from the manner in which the concept of function is articulated in the several fundamental functions.

The single functions $a \log x, a^x, x^a, ax, a + x$ can, namely, be derived in such a way

⁸By its formulas mathematics is just fitted to express the real laws. If, in the determination of these laws, it merely came to naming the universal, they could be determined without the help of mathematics; but since it also comes to the phenomena and the individual cases, inasmuch as the laws are the laws of the phenomena, the mathematical must come in, that the expression may become exact. He who, for example, grasps the law of uniformly increasing velocity only in so far as it is logically expressed in words, grasps indeed in a certain sense the essential, but, since he lacks the punctuality in the particular cases, not the real. — It makes a great difference whether the law for the collision of hard bodies is set forth in logical form or in mathematical formula. The logical presentation in words is, with all its accuracy, always hovering; only a formula such as $\frac{MH+mh}{M+m}$ is fitted to explain the single.

that the succeeding one springs with inner necessity from the preceding. However much difference there may be here between these functions in their independent appearance, one nevertheless easily finds, on the thread of development, the knots where the one lies enclosed in the other.

Here, thus, is a turning-point, where the concept of multiplication dawns beneath the veil of addition. If one takes $a + a + a$, a sum of three equal addends, one comes almost unawares to conceive the magnitude as an a repeated 3 times. But therewith the transition is not yet made. Consciousness can often come very near to a new concept, can glimpse it as in a dawn, and yet the concept can again vanish and be lost. If a new concept is to arise for cognition, it is not enough that the investigation leads up to the turning-point where it has its origin (for then the great discoveries of the present would surely have been made centuries ago); in order that a new concept may come to be in cognition, it is required that thought rightly catch sight of it, take it up into a category, introduce it among the other concepts, follow its course, and test its consequences.

As it goes in mathematics with the transition from addition to multiplication, so also with the transition from multiplication to the exponential function, and from this to the logarithm⁹.

In the equation $a + a + a + \dots a = na$ the same magnitude is conceived on the one side as sum, on the other side as product. Here too it is expressly seen that the mathematical notation is a language.

As an immediate expression of calculation the equation indicates only that on each side of the equals-sign there are equal magnitudes; but with respect to the concept the equation is a judgment.

Seen from the logical side, this judgment again shows the contradiction, the contradiction that multiplication is thus both the same and yet not the same as addition. This contradiction, together with the *diversus respectus* by which formal logic would avoid the contradiction, and just thereby comes into contradiction with itself, and falls under dialectics, does not concern mathematics at all.

As a mathematical judgment the equation will not logically declare what multiplication either is or is not, but rather demonstrate, in the exact form, how the

⁹The inner connection of the trigonometric functions with the exponential and with the logarithm can only be pointed out later.

intellectual action of multiplication is connected with that of addition. What dialectics is in philosophy, the intellectual action with its practical transitions is in mathematics. That the matter stands thus, we can illustrate by a retrospect.

The mathematical function presupposes the variable in its dependents as in its independents; only by grasping the infinite possibility of the changes does one grasp the function. But the function presupposes, in all these changes, at the same time an invariable; it expresses, after all, the law, and the law is the invariable.

Seen from the finite side, it might well seem as if the invariable in the formula were a dead schema, indifferent toward the variable cases — that, for example, with respect to the formula $x = \frac{1}{\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots \frac{1}{a_n}}$, it could be quite indifferent what values one inserted. This is indeed, in the finite sense, so. But the more finite the point of view becomes, the more the concept of function also recedes into the background; the more, on the contrary, the concept of function prevails, the more intimately is seen the unity of the variable and the invariable: the function points precisely to that which dialectics demands.

In the above problem the times $a_1, a_2 \dots a_n$ were given, each for itself, and consequently independent of one another. But one might assume a_1 given in such a way that $a_2, a_3 \dots$ were made dependent on it, and that according to such a law that they formed, for example, either an arithmetic or a geometric series: $a_1 = a_1$, $a_2 = a_1 + d$, $a_3 = a_1 + 2d \dots$ or $a_1 = a_1$, $a_2 = a_1 q$, $a_3 = a_1 q^2, \dots$ thus in the first case $a_p = a_1 + (p-1)d = \varphi_1(p)$; in the second case $a_p = a_1 q^{p-1} = \varphi_2(p)$; we then obtained

$$x = \frac{1}{\frac{1}{\varphi(1)} + \frac{1}{\varphi(2)} + \dots \frac{1}{\varphi(p)} \dots \frac{1}{\varphi(n)}} = F(\varphi(p)) = f(p).$$

Since in the expression for x there is found in the denominator a sum of the form $\frac{1}{\varphi(p)}$, which is a function of $\varphi(p)$, and the whole sum consequently a function of $\varphi(p)$, so also 1, divided by the sum, is a function of $\varphi(p)$. The invariable and the variable will in this way interpenetrate one another.

The mathematical functions are not dead formulas, but active, busy, practical concepts that lead an intellectual life in the operations of science. For such practical transitions no dialectical spur is needed.

As an example of how the mathematical concepts of action can cooperate even

within elementary limits, we will, in conclusion, recall the symmetric functions.

Through

$$\begin{aligned}(x - a_1)(x - a_2) &= x^2 - (a_1 + a_2)x + a_1 a_2 = x^2 + A_1 x + A_2; \\(x - a_1)(x - a_2)(x - a_3) &= x^3 - (a_1 + a_2 + a_3)x^2 + (a_1 a_2 + a_1 a_3 + a_2 a_3)x - a_1 a_2 a_3 \\&= x^3 + A_1 x^2 + A_2 x + A_3,\end{aligned}$$

one will understand the general expression:

$$f(x) = x^n + A_1 x^{n-1} + A_2 x^{n-2} + A_3 x^{n-3} + \dots + A_{n-1} x + A_n = (x - a_1)(x - a_2)(x - a_3) \dots (x - a_n) = 0.$$

Here the invariable is everywhere present in the variable; for the law of development shows us the reciprocal determination of coefficients and roots:

$A_1 = -(a_1 + a_2 + a_3 + \dots a_n) = f_1(a_1 a_2 \dots a_n)$ = the sum of all the a 's with opposite sign; here the first fundamental function operates.

$A_3 = -(a_1 a_2 a_3 + a_1 a_2 a_4 + \dots a_1 a_2 a_n + a_2 a_3 a_4 + \dots a_{n-2} a_{n-1} a_n) = f_3(a_1 a_2 a \dots a_n)$ = the sum of the products of the a 's taken three at a time, with opposite sign; here the first and second fundamental functions cooperate. $A_n = (-1)^n(a_1 a_2 a_3 \dots a_n) = f_n(a_1, a_2, \dots a_n)$ = the product of all the a 's with opposite sign, when n is odd. It is thus the second fundamental function that here appears.

If one considers besides that the series proceeds according to powers of x , whereby the alternation of the exponential function with the logarithmic also becomes possible, then one obtains a total impression of the intellectual cooperation of the functions.

The formal concepts of logic are like empty vessels, which must either remain in their emptiness or be filled with matter and content from without; the function-concepts of mathematics are like active organs that assimilate to themselves an essential content.

Although the mathematical operations, in their peculiar procedure, belong to the specialized science without concerning philosophy, their character nevertheless lends itself to a philosophical consideration, to illuminate the activity of thought from a side of its own, and to make visible problems whose treatment, not to say solution, is and must be the aim of philosophy.